

Merewether High School
Trial Higher School Certificate Examination 1999
Mathematics 3U Additional, 4U Common Paper

Time Allowed: 2 hours, plus 5 minutes reading time.

Instructions:

- Marks may be deducted for untidy or poorly arranged work.
- Approved calculators may be used, and the table of standard integrals is provided.
- Show all necessary working - a correct answer without appropriate working may not receive full marks.
- Put your student's number on each page.
- START EACH QUESTION ON A NEW SHEET OF PAPER.

QUESTION 1 (12 marks)

(a) Solve for $0 \leq x \leq 2\pi$

$$\cos 2x - 3\sin x - 2 = 0 \quad 3$$

(b) Find $\int x\sqrt{1-x} \, dx$, using the substitution $u = 1 - x$ 3

(c)

(i) Sketch the graph of the function

$$y = 2\tan^{-1}x \quad 2$$

(ii) What value does $2\tan^{-1}x$ approach as x increases indefinitely? 1

(iii) Find the exact equation of the tangent to the curve

$$y = 2\tan^{-1}x, \text{ at the point where } x = 1. \quad 3$$

QUESTION 2 (12 marks)

(a) Solve $x + \frac{1}{x} \geq 2$

2

(b) Find:

(i) $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$

2

(ii) $\int \frac{x dx}{x^2 + 1}$

2

(c) If α , β and γ are the roots of the equation

$$2x^3 + 6x - 3 = 0,$$

find the value of:

(i) $\alpha + \beta + \gamma$, $\alpha\beta + \alpha\gamma + \beta\gamma$ and $\alpha\beta\gamma$

3

(ii) $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$

3

QUESTION 3 (12 marks)

(a) Find the derivative, with respect to x , of

(i) $\log_e(\cos^{-1}x)$

2

(ii) $\sin^{-1}5x$

2

(b) Use Mathematical Induction to prove that for all positive integral values of n :

$$1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + n \times (n + 1) = \frac{n(n+1)(n+2)}{3}$$

5

QUESTION 3 continued

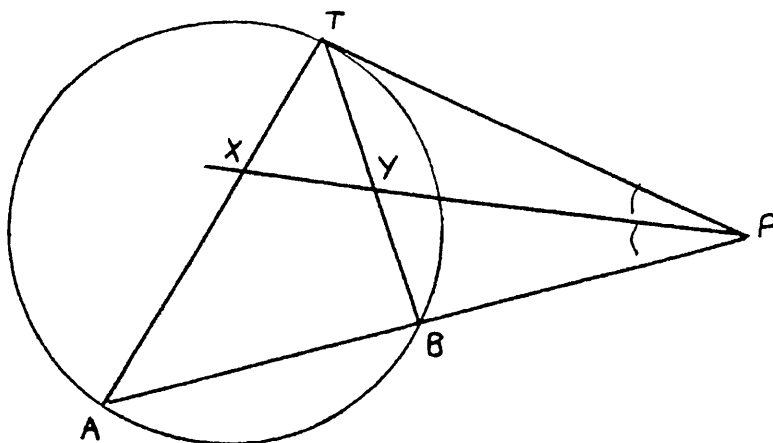
- (b) The area under the curve $y = \sin x$ between $x = 0$ and $x = \frac{\pi}{4}$ is rotated about the x-axis through one complete revolution. Find the volume of the solid formed, to one decimal place. 3

QUESTION 4 (12 marks)

- (a) Find the acute angle between the straight lines $2y - x + 1 = 0$ and $y = 5x + 2$, giving the answer correct to the nearest degree. 2
- (b) A is the point $(-2, -1)$ and B is the point $(1, 5)$. Find the co-ordinates of the point Q, which divides the interval AB externally in the ratio 5:3 2
- (c) (i) Express $7\cos\theta - \sin\theta$ in the form $R\cos(\theta + \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$ 2
- (ii) Hence solve $7\cos\theta - \sin\theta = 5$ for $0^\circ \leq \theta \leq 360^\circ$, giving answers to the nearest degree. 3
- (c) Evaluate exactly $\int_0^1 (e^{-x} + \frac{1}{1+x} - \frac{1}{\sqrt{1-x^2}}) dx$ 3

QUESTION 5 (12 marks)

(a)



The tangent at T on the circle meets a chord AB produced at P. The bisector of $\angle TPA$ meets TA and TB at X and Y respectively.

- | | |
|---|---|
| (i) Give the reason why $\angle PTB = \angle TAB$ | 1 |
| (ii) Prove that $TX = TY$. | 3 |
| (iii) Prove that $\frac{TX}{XA} = \frac{TY}{PA}$ | 2 |

(b) In how many ways can a train of nine carriages be arranged if four of the carriages (A, B, C, D):

- | | |
|--|---|
| (i) are to be at the rear in the order ABCD with D last? | 1 |
| (ii) must be kept together but in any order? | 2 |

(c) (i) Expand $\tan(\alpha + \beta)$

1

(ii) If $\tan A$ and $\tan B$ are the roots of the equation

$$3x^2 - 5x - 2 = 0,$$

find the value of $\tan(A + B)$.

2

QUESTION 6 (12 marks)

- (a) The tangent at P ($2ap, ap^2$) on the parabola $x^2 = 4ay$ meets the x-axis in T. The normal at P meets the y-axis in N.

- (i) Find the co-ordinates of M, the midpoint of TN. The equations of the tangent and the normal need NOT be derived.

2

- (ii) Show that the locus of M is the parabola

$$x^2 = \frac{a}{2}(y - a).$$

3

- (b) A circular oil slick lies on the surface of calm water. Its area is increasing at the rate of $12 \text{ m}^2/\text{min}$. At what rate is the radius increasing at the time at which the radius is 3 metres?

3

- (c) Find $\frac{d}{dx}(\sqrt{1-x^2} + x \sin^{-1} x)$. Hence evaluate $\int_0^{\frac{1}{2}} \sin^{-1} x \, dx$ correct to three significant figures.

4

QUESTION 7 (12 marks)

- (a) A certain particle moves along the x-axis according to the law

$$t = 2x^2 - 5x + 3$$

where x is measured in centimetres and t in seconds. Initially the particle is 1.5 cm to the right of the origin O and moving away from O.

- (i) Prove that the velocity, $v \text{ cms}^{-1}$, is given by

$$v = \frac{1}{4x - 5}$$

2

- (ii) Find an expression for the acceleration, $a \text{ cms}^{-2}$, in terms of x .

2

- (iii) Find the velocity of the particle when $t = 6$ seconds.

3

QUESTION 7 continued

(d) A particle is moving in Simple Harmonic Motion with acceleration

$$\ddot{x} = -4x \text{ mms}^{-2}$$

If the particle starts at the origin with a velocity of 8 mms^{-1} .

(i) State its period

1

(i) Find its displacement after $\frac{\pi}{3}$ seconds.

4

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1. a) $\cos 2x - 3\sin x - 2 = 0$

$$(1 - 2\sin^2 x) - 3\sin x - 2 = 0$$

$$2\sin^2 x + 3\sin x + 1 = 0$$

$$(2\sin x + 1)(\sin x + 1) = 0$$

$$\sin x = -\frac{1}{2}, -1$$

$$\therefore x = \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{3\pi}{2}$$

b) $I = \int x\sqrt{1-x} \cdot dx$

let $u = 1-x \Rightarrow x = 1-u$

$$\therefore du = -1 \cdot dx$$

$$\therefore I = \int (1-u)\sqrt{u} \cdot -du$$

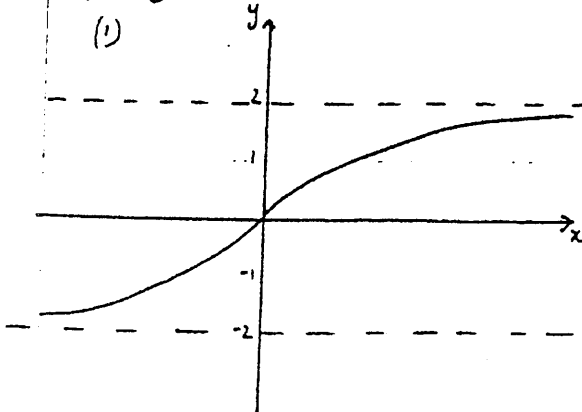
$$= \int (u^{\frac{3}{2}} - u^{\frac{1}{2}}) du$$

$$= \frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} + C$$

$$= \frac{2(1-x)^{\frac{5}{2}}}{5} - \frac{2(1-x)^{\frac{3}{2}}}{3} + C$$

c) $y = 2\tan^{-1}x$

(i)



(ii) as $x \rightarrow \infty$, $2\tan^{-1}x \rightarrow 2$

(iii) at $x = 1$

$$y = 2\tan^{-1}1$$

$$= \frac{\pi}{2}$$

also $\frac{dy}{dx} = \frac{2}{1+x^2}$

at $x=1$, $\frac{dy}{dx} = \frac{2}{1+1} = 1$

eqn of Tangent:

$$\frac{y-y_1}{x-x_1} = m$$

$$\frac{y-\frac{\pi}{2}}{x-1} = 1$$

$$y - \frac{\pi}{2} = x - 1$$

$$\therefore y = x - 1 + \frac{\pi}{2} \text{ is eqn of tangent}$$

2 a) $x + \frac{1}{x} \geq 2$

$$x \neq 0$$

Solve $x + \frac{1}{x} = 2$

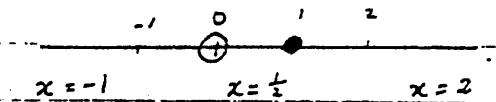
$$x^2 + 1 = 2x$$

$$x^2 - 2x + 1 = 0$$

$$(x-1)^2 = 0$$

$$x = 1$$

Graph + test



$$-1 + \frac{1}{-1} \geq 2$$

False

$$\frac{1}{2} + \frac{1}{\frac{1}{2}} \geq 2$$

True

$$2 + \frac{1}{2} \geq 2$$

True

$$\therefore x > 0 \text{ is the solution}$$

b) (i) $I = \int_0^1 \frac{dx}{\sqrt{1-x^2}}$

$$= [\sin^{-1}x]_0^1$$

$$= \sin^{-1}1 - \sin^{-1}0$$

$$= \frac{\pi}{2}$$

(ii) $I = \int \frac{x}{x^2+1} \cdot dx$

$$= \frac{1}{2} \int \frac{2x}{x^2+1} dx$$

$$= \frac{1}{2} \ln(x^2+1) + C$$

$$c) 2x^3 + 0x^2 + 6x - 3 = 0$$

$$(i) \alpha + \beta + \gamma = -\frac{b}{a} = -\frac{0}{2} = 0$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = \frac{-3}{2} = 3$$

$$\alpha\beta\gamma = -\frac{d}{a} = -\frac{-3}{2} = 1\frac{1}{2}$$

$$(ii) \text{Exp} = \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

$$= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$= \frac{3}{1\frac{1}{2}}$$

$$= 2$$

$$3. a) (i) f(x) = \ln(\cos^{-1}x)$$

$$f'(x) = \frac{1}{\cos^{-1}x} \times \frac{-1}{\sqrt{1-x^2}}$$

$$= \frac{-1}{\cos^{-1}x \cdot \sqrt{1-x^2}}$$

$$(ii) f(x) = \sin^{-1}5x$$

$$f'(x) = \frac{1}{\sqrt{1-25x^2}} \times 5$$

$$= \frac{5}{\sqrt{1-25x^2}}$$

b) To prove $S(n)$:

$$1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$$

Step 1 Test $S(1)$

$$\text{LHS} = 1 \times 2 = 2$$

$$\text{RHS} = \frac{1(1+1)(1+2)}{3} = \frac{6}{3} = 2$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore$ statement is true for $n=1$.

Step 2: Assume $S(k)$ is true

$$\text{i.e. } 1 \times 2 + 2 \times 3 + \dots + k(k+1) = \frac{k(k+1)(k+2)}{3}$$

Step 3: Test $S(k+1)$

$$\text{i.e. Is } 1 \times 2 + 2 \times 3 + \dots + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$$

$$\text{LHS} = 1 \times 2 + 2 \times 3 + \dots + k(k+1) + (k+1)(k+2)$$

$$= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2), \text{ from (2)}$$

$$= \frac{k(k+1)(k+2) + 3(k+1)(k+2)}{3}$$

$$= \frac{(k+1)(k+2)[k+3]}{3}$$

$$= \text{RHS}$$

$\therefore$ statement is true for $n=k+1$

if it is true for $n=k$.

Step 4 $S(1)$ is true

$\therefore S(1+1)$ is true i.e. $S(2)$ is true

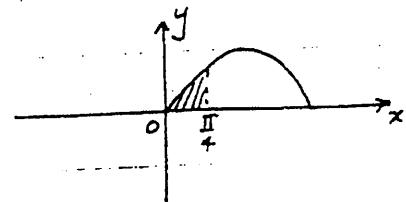
$\therefore S(2+1)$ is true i.e. $S(3)$ is true

and so on.

$\therefore$ by the principle of Maths.

Induction $S(n)$ is true for all positive integral values of n .

b)



$$V = \pi \int_a^b y^2 dx$$

$$= \pi \int_0^{\pi/4} \sin^2 x dx$$

$$= \pi \int_0^{\pi/4} \sin^2 x dx$$

$$\begin{aligned}
 &= \pi \int_0^{\frac{\pi}{4}} \frac{1}{2} (1 - \cos 2x) dx \\
 &= \frac{\pi}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\frac{\pi}{4}} \\
 &= \frac{\pi}{2} \left[\frac{\pi}{4} - \frac{\sin \frac{\pi}{2}}{2} - 0 + \frac{\sin 0}{2} \right] \\
 &= \frac{\pi}{2} \left[\frac{\pi}{4} - \frac{1}{2} \right] \\
 &= 0.05 \text{ units}^3 \quad (2 \text{ dec. pl.})
 \end{aligned}$$

4. a) $y = \frac{1}{2}x - \frac{1}{2}$

2 $\therefore m_1 = \frac{1}{2}$

$y = 5x + 2$

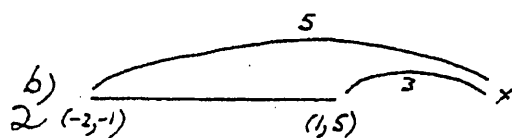
$\therefore m_2 = 5$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$= \left| \frac{\frac{1}{2} - 5}{1 + \frac{1}{2} \times 5} \right|$

$= 1 \frac{2}{7}$

$\therefore \theta = 52^\circ \text{ (nearest degree)}$



$x = \frac{kx_2 + lx_1}{k+l}$

$= \frac{5 \times 1 + -3 \times -2}{5 + -3}$

$= 5 \frac{1}{2}$

$y = \frac{ky_2 + ly_1}{k+l}$

$= \frac{5 \times 5 + -3 \times -1}{5 + -3}$

$= 14$

$\therefore \phi \text{ is } (5 \frac{1}{2}, 14)$

c) (1) let $7 \cos \theta - \sin \theta = R \cos(\theta + \alpha)$
2 $= R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$

$\therefore R \cos \alpha = 7$

$R \sin \alpha = 1$

$\therefore R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 49 + 1$

$R = \sqrt{50}, R > 0$

$= 5\sqrt{2}$

also $\frac{R \sin \alpha}{R \cos \alpha} = \frac{1}{7}$

$\tan \alpha = \frac{1}{7}$

$\alpha = 8^\circ 8'$

$\therefore 7 \cos \theta - \sin \theta = 5\sqrt{2} \cos(\theta + 8^\circ 8')$

(11) $7 \cos \theta - \sin \theta = 5$

3 $5\sqrt{2} \cos(\theta + 8^\circ 8') = 5$

$\cos(\theta + 8^\circ 8') = \frac{1}{\sqrt{2}}$

$\therefore (\theta + 8^\circ 8') = 45^\circ, 315^\circ$

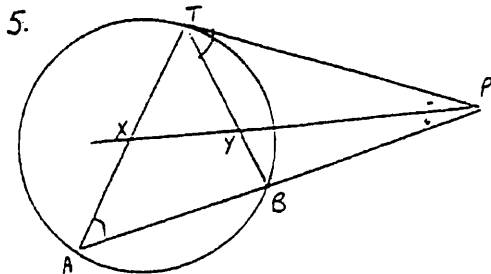
$\theta = 37^\circ, 307^\circ \text{ (nearest deg)}$

d) $I = \int_0^1 \left(e^{-x} + \frac{1}{1+x} - \frac{1}{\sqrt{1-x^2}} \right) dx$

3 $= \left[-e^{-x} + \ln(1+x) - \sin^{-1} x \right]_0^1$

$= (-e^{-1} + \ln 2 - \sin^{-1} \frac{1}{2}) - (-e^0 + \ln 1 - \sin^{-1} 0)$

$= -\frac{1}{e} + \ln 2 - \frac{\pi}{2} + 1$



(i) The angle between a tangent and a chord of contact is equal to the angle in the alternate segment

(ii) In Δ 's AXP and TYP ,

$\angle PTB = \angle TAB$ from (i)

$\angle TPY = \angle XPA$ (PX bisector of $\angle TPA$, data)

$\therefore \Delta AXP \sim \Delta TYP$ (2 pts corr. $\angle$'s equal)

$\therefore \angle AXP = \angle TYP$ (3rd $\angle$ sim Δ 's)

$\angle TXY = \angle 180^\circ - \angle AXP$ ($\angle$'s st. line)

and $\angle TYX = \angle 180^\circ - \angle TYP$ ($\angle$'s st. line)

$\therefore \angle TXY = \angle TYX$

$\therefore \Delta TXY$ is isosceles (base $\angle$'s equal)

$\therefore TX = TY$ (equal sides isos. Δ)

(iii) Since Δ 's AXP & TYP are similar from (ii),

Corr. sides are in same ratio

$$\therefore \frac{TY}{XA} = \frac{TP}{PA}$$

But $TY = TX$ from (ii)

$$\therefore \frac{TX}{XA} = \frac{TP}{PA}$$

b) (i) ABCD is 1 arrangement.

$\therefore$ ABCD are at the end + 5 other canoes must be arrange at the front.

$$\begin{aligned} \text{No of ways} &= 5! \times 1 \\ &= 120 \end{aligned}$$

(ii) A, B, C and D can be arranged in $4!$ ways. 2

$\therefore$ there are 6 'objects' to be arrange

$$\begin{aligned} \text{No of ways} &= 6! \times 4! & 2880 \\ &= 17,280 & 5! \times 4! \end{aligned}$$

Page 1

c) (i) $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$

(ii) $3x^2 - 5x - 2 = 0$

$(x-2)(3x+1) = 0$ 2

$x = 2, -\frac{1}{3}$ 1

$\therefore$ let $\tan A$ be 2, $\tan B$ be $-\frac{1}{3}$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

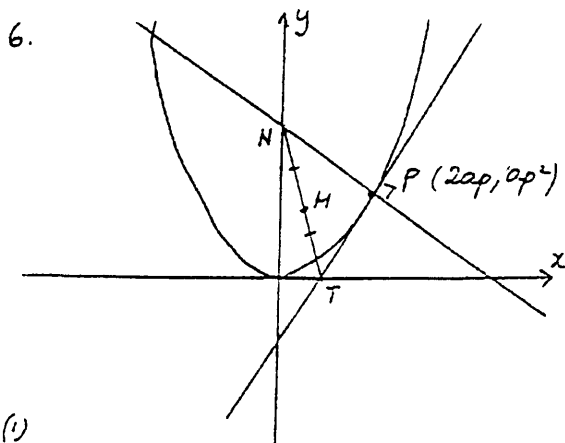
$$= \frac{2 + (-\frac{1}{3})}{1 - 2 \times (-\frac{1}{3})}$$

$$= \frac{1\frac{2}{3}}{\frac{5}{3}} = 1\frac{2}{5}$$

$$= 1$$

Note same answer if $\tan A$ is $-\frac{1}{3}$, $\tan B$ is 2.

$$\begin{aligned} \text{or } \tan A + \tan B &= \frac{5}{3} \\ \tan A \tan B &= -\frac{2}{3} \end{aligned} \quad \left. \vphantom{\begin{aligned} \tan A + \tan B &= \frac{5}{3} \\ \tan A \tan B &= -\frac{2}{3} \end{aligned}} \right\} 1$$



(i)

Tangent at P is $y - px + ap^2 = 0$

let $y = 0$ ie $-px + ap^2 = 0$

$$x = ap$$

$\therefore T$ is $(ap, 0)$

normal at P is $x + py = ap^3 + 2ap$

let $x = 0$ ie $py = ap^3 + 2ap$

$$y = ap^2 + 2a$$

$\therefore N$ is $(0, ap^2 + 2a)$

$\therefore M$ is $(\frac{ap+0}{2}, \frac{0+ap^2+2a}{2})$

$$\text{ie } (\frac{ap}{2}, \frac{ap^2+2a}{2})$$

(ii) parametric equations of locus of M

$$\text{or } x = \frac{ap}{2} \quad (i)$$

$$y = \frac{ap^2+2a}{2} \quad (ii)$$

$$\text{from (i)} \quad p = \frac{2x}{a}$$

sub. into (ii)

$$y = \frac{a(\frac{2x}{a})^2 + 2a}{2}$$

$$2y = \frac{4ax^2}{a^2} + 2a$$

$$2y = \frac{4x^2}{a} + 2a$$

$$2ay = 4x^2 + 2a^2$$

$$\therefore 4x^2 = 2ay - 2a^2$$

$$+x^2 = 2a(y-a)$$

$$x^2 = \frac{2a}{4}(y-a)$$

$\therefore x^2 = \frac{a}{2}(y-a)$ is Cartesian equation of the locus of M.

$$b) A = \pi r^2$$

$$\therefore \frac{dA}{dr} = 2\pi r$$

given that $\frac{dA}{dt} = 12$, find $\frac{dr}{dt}$

$$\frac{dr}{dt} = \frac{dr}{dA} \times \frac{dA}{dt} \text{ by the Chain Rule}$$

$$= \frac{1}{2\pi r} \times 12$$

at $r = 3$

$$\frac{dr}{dt} = \frac{1}{2\pi(3)} \times 12$$

$$= \frac{2}{\pi}$$

$\therefore$ radius is increasing at $\frac{2}{\pi}$ m/min

$$c) \text{Exp} = \frac{d}{dx} (\sqrt{1-x^2} + x \sin^{-1}x)$$

$$= \frac{1}{2}(1-x^2)^{-\frac{1}{2}} - 2x + \sin^{-1}x + x \times \frac{1}{\sqrt{1-x^2}}$$

$$= \frac{-x}{\sqrt{1-x^2}} + \sin^{-1}x + \frac{x}{\sqrt{1-x^2}}$$

$$= \sin^{-1}x$$

$$\therefore I = \int_0^{\frac{\pi}{2}} \sin^{-1}x \, dx$$

$$= \left[\sqrt{1-x^2} + x \sin^{-1}x \right]_0^{\frac{1}{2}}$$

$$= \left(\sqrt{\frac{3}{4}} + \frac{1}{2} \sin^{-1}\frac{1}{2} \right) - (1 + 0 \sin^{-1}0)$$

$$= \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{\pi}{6} - 1$$

$$= \frac{\sqrt{3}}{2} + \frac{\pi}{12} - 1$$

$$= 0.128 \text{ to 3 sig. fig.}$$

7. a) $L = 2x^2 - 5x + 3$

(i) $\frac{dt}{dx} = 4x - 5$

$\therefore \frac{dx}{dt} = \frac{1}{4x-5}$

$\therefore v = \frac{1}{4x-5}$

(ii) $\frac{d^2x}{dt^2} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$= \frac{d}{dx} \left(\frac{1}{2} \cdot \frac{1}{(4x-5)^2} \right)$

$= \frac{1}{2} \times -2(4x-5)^{-3} \times 4$

$= \frac{-4}{(4x-5)^3}$

$\therefore a = \frac{-4}{(4x-5)^3}$

(iii) at $L=6$, $6 = 2x^2 - 5x + 3$

$2x^2 - 5x - 3 = 0$

$(x-3)(2x+1) = 0$

$x = 3, -\frac{1}{2}$

But $\frac{1}{4x-5} \neq 0$

$\therefore v \neq 0$

$\therefore$ particle doesn't change direction
it starts at $x=1.5$ & moves right

$\therefore$ at $L=6$, $x=3$

at $x=3$ $v = \frac{1}{4(3)-5}$

$= \frac{1}{7}$

$\therefore$ velocity is $\frac{1}{7} \text{ cm.s}^{-1}$ when $t=6 \text{ s}$.

7. d) $\ddot{x} = -4x$

$\therefore n = 2$

(i) period $= \frac{2\pi}{n}$
 $= \frac{2\pi}{2}$
 $= \pi$

(ii) $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$\therefore \frac{1}{2} v^2 = \int -4x dx$
 $= -2x^2 + C$

at $x=0$, $v=8$

$32 = 0 + C$

$C = 32$

$\therefore \frac{1}{2} v^2 = -2x^2 + 32$

$v^2 = 64 - 4x^2$

$\therefore v = \pm \sqrt{64 - 4x^2}$

$= \pm 2\sqrt{16 - x^2}$

at $x=0$, v is positive

$\therefore v = 2\sqrt{16 - x^2}$

$\frac{dx}{dt} = 2\sqrt{16 - x^2}$

$\frac{dt}{dx} = \frac{1}{2} \cdot \frac{1}{\sqrt{16 - x^2}}$

$\therefore t = \frac{1}{2} \sin^{-1} \frac{x}{4} + k$

at $L=0$, $x=0$

$\therefore 0 = 0 + k$

$\therefore t = \frac{1}{2} \sin^{-1} \frac{x}{4}$

$2t = \sin^{-1} \frac{x}{4}$

$\therefore \frac{x}{4} = \sin 2t$

$x = 4 \sin 2t$

at $t = \frac{\pi}{3}$

$x = 4 \sin \frac{2\pi}{3}$

$= 4 \times \frac{\sqrt{3}}{2} = 2\sqrt{3}$

$\therefore$ displacement is $2\sqrt{3} \text{ mm}$ after $\frac{\pi}{3} \text{ s}$